NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

# R23.4b Convection-scale AI and thunderstorm nowcasting

**Group** AI Weather & Climate Models for Wind Engineering

**Status** Shelved

**Period** 2026-10-14 14:21 — 2026-10-16 11:59

## Stage conclusion

## Project report

# Convection-Allowing AI and Nowcasting (L / M / L2)

## Objective

Determine whether ~3 km convection-allowing AI models (StormCast, WoFSCast, etc.) can directly predict thunderstorms and surface wind gusts, and whether they can perform batch forecasts/hindcasts across tens to hundreds of events—which is the only scalable route for the thunderstorm direction (Liang Hu, 10-14). Step 1 deliverables: a verification table of whether publicly available models can run on Colab A100; a verification table comparing 0–6 h lead-time forecasts against ASOS observations and HRRR across 5–10 downburst/severe thunderstorm events with station measurements; and AI/HRRR soundings vs. observed soundings on the same batch of events (prerequisite gate for L2).

# Current Status

10-16 **Shelved** (5/6 valid, verdict). What is shelved is the sample expansion of the configuration "StormCast V1/CONUS + 10 m wind × 1.6677 gust proxy + retrospective ERA5 conditioning": under the frozen thresholds, neither generation met the sample expansion criteria (V1: 20 events; CONUS: 12 events, 53/54 groups). This is **not** a claim that "convection-allowing AI is inferior to HRRR": hit-rate differences are not statistically significant (McNemar p = 0.22–1.0), Criterion 1 passes across all lead times if switched to the ensemble-maximum definition, any-member ≥25.7 m/s counts for AI are at least as high as HRRR GUST (V1: 9/11/9 vs. 8/8/6), V1 includes 7 previously seen cases, and gusts used only a constant scaling factor. Why not continue: the current experimental design can no longer answer whether scaling up is worthwhile, and further sample expansion would merely spend compute on a non-significant gap (CONUS costs ~700 CU per 100 events). Restart conditions (if any is met, open a new topic and pre-register new criteria rather than directly resuming sample expansion): ① blind sample + thunderstorm wind verification + convective non-severe-wind controls + pre-registered ensemble aggregation and probabilistic criteria + sample size estimation; ② mixed-layer wind profile gust diagnostics replacing the constant factor; ③ availability of a public convection-allowing model with native gust output or out-of-training-period verifiability; ④ changing the target to a batch application that does not require station peak accuracy (change of objective, subject to Liang Hu's decision). Round 2 used ~212 CU in total. M is closed; L2 was not started.

# Round 2 Criteria (Frozen on 10-14; No Post-Hoc Relaxation)

See the "Criteria" section: across each 1–3 h lead time, median peak ratio of AI gust estimates ≥ 0.9× HRRR GUST; for ≥25.7 m/s detection (flagged when ensemble P ≥ 0.3) on combined event + non-event samples, hit rate not lower than HRRR GUST and false alarm rate not exceeding 1.5× HRRR GUST; GFS real-forecast conditioning degradation relative to ERA5 not exceeding 20%. Gust factor fixed at 1.6677 (fitted on 2022). Wide and narrow windows reported side by side, window-edge fraction reported, two HRRR baselines; persistence included only in the appendix.

# Step 1 Criteria (Fixed Before Execution)

- Verification target: for each event, take the station with the peak ASOS observed wind; for the model, take the maximum 10 m wind (and gust variable if available) within ±25 km centered on the station and ±1 h around the observed peak time. Baselines: operational HRRR forecasts at the same lead time (10 m wind and GUST), plus persistence of the HRRR analysis at initialization time.
- Metrics: peak ratio (model peak / observed peak), peak timing and location errors, and whether ≥25.7 m/s (50 kt) is predicted.
- "Worth continuing": across 1–6 h lead times, AI model peak ratio ≥ 90% of HRRR, outperforming persistence on at least 2/3 of events; and single-event single-initialization inference time ≤10 GPU minutes (prerequisite for scaling up). If unmet, document exactly where it falls short without relaxing the criteria.

- Events must lie outside the training period of the weights used: StormCast V1 (trained 2018-07 to 2021-12, Apache-2.0 weights) uses post-2022 events; StormCast-CONUS (trained through 2025-09-30, OpenMDW-1.1 weights) uses post-2025-10 events.

## Path to Scaling Up

Write the Step 1 scripts as an automated batch pipeline ("event list → automatic HRRR initialization retrieval → inference → automatic verification against ASOS"); once past the gate, expand to tens to hundreds of thunderstorm high-wind events with ASOS observations in SPC/Storm Events for hindcast statistics; AIWP statistics from M establish the predictable lead time of the environment, determining whether L2 can perform "forecasts" or only "hindcasts." The CM1 ensemble in L2 runs on the supercomputer, with full-scale production handed over to the CM1 line or promoted to a formal project.

## Boundaries

- Hierarchy (Root Lead 10-14, 5-level cap): This node is a Level-2 node and will not open Level-3 sub-projects; when subdivision is needed, use tasks within this node or report to the parent node Lead (Thunderstorms and Convective Scale) to open a peer Level-2 node.
- Only two hops remain downward: this node → bot (`ideas`) → tool; task briefs must not allow tool bots (`gpu-gemini`, reviewer, `litwiki`, etc.) to further delegate to others.
- Subtree lists are not grouped by parent node; the UI project tree is authoritative for hierarchy.
- Compute and data: Colab prioritized, with no single-experiment cap (aligned by the parent project on 10-14 per Liang Hu's verbatim instruction "use Colab heavily at this stage without CU limits"; the previous "≤20 CU per experiment" cap is rescinded); pause and report to the parent Lead if a single experiment is estimated to exceed 100 CU; must run `colab stop` when finished so sessions never idle; dispatch only one executor per round. Round 2 of L remains under its existing cap: ≤200 CU for the entire round, report to the parent Lead once cumulative usage reaches 120 CU, and stop and report at 200 CU. Supercomputer single jobs ≤50 SU via resource request; L2 CM1 ensembles are run by the CM1 line, while we only deliver profiles in `input_sounding` format.
- Compute and storage locations (tightened by Liang Hu on 10-14): the workstation does not run compute or store compute data (46 GB free on the C drive as of 10-15 01:59). All solvers, inference, training, batch cases, and large downloads must run and be stored on Colab or CRC/TACC; the workstation holds only scripts, small results <200 MB per item, figures, and reports, keeping retrieved checkpoints as small files while leaving large intermediates on Colab or the cluster; if >1 GB must be temporarily placed on the workstation, report to the parent Lead first. CRC group space is ~95% full, so new large datasets should prioritize TACC work or scratch storage; CRC defaults to a single node. Every task brief must explicitly specify where computation runs and where outputs are stored.

# Settled Decisions

- 2026-10-14 Literature review verdict: separate the two StormCast weight versions; reclassify FuXi-Nowcast as a counterexample (inferior to persistence at severe gust thresholds); add HRRRCast as a candidate (public version is 6 km, GUST is a post-processed diagnostic, pressure levels up to 200 hPa, GFS boundaries) and retract "only StormCast is usable"; narrow the framing of the Cook 10% finding; retract the Collins extrapolation; state novelty under a narrow framing →
- 2026-10-14 Opened; Step 1 entirely on Colab and the workstation, consult the CM1 line first on the CM1 portion →

# Trials and Results

| Date | Action Taken | One-Line Result |
|---|---|---|
| 10-16 | Round 2 CONUS segment (B: 46.2 + B2: minimal + B3: 78.1 CU, estimated): 12 events in 2026 + 6 controls, 53/54 groups (missing 1 member in one 3 h control group) | **Failed under frozen criteria**. Wide-window AI/obs at 1/2/3 h: 0.67/0.74/0.68, HRRR GUST: 0.79/0.82/0.81, HRRR10×factor: 0.84/0.94/0.86; AI/HRRR GUST: 0.85/0.90/0.84 → Criterion 1 passes only at 2 h; hits AI 1/1/0 vs. HRRR GUST 4/2/3 (/12) → Criterion 2 fails; false alarms 0 for both. The missing control group affects only false alarms and does not alter the conclusion |
| 10-15 | Round 2 Segment A: full V1+ERA5 sample of 20 events + 10 non-event controls × 1/2/3 h × 10 members (~25.7 CU in this session, burn-rate estimate) | **Failed under frozen criteria**. Wide-window peak ratios AI / HRRR GUST / HRRR10×factor: 1 h 0.68/0.73/0.90, 2 h 0.68/0.71/0.75, 3 h 0.64/0.75/0.85 → Criterion 1 (AI ≥ 0.9×HRRR GUST) passes at 1 h (0.93) and 2 h (0.96), fails at 3 h (0.85); Criterion 2 hits AI 4/5/4 vs. HRRR GUST 8/8/6 (/20), fails; control false alarms 0/10 for both. Even HRRR 10 m wind scaled by the same factor mostly outperforms AI. Narrow window reaches only 0.4–0.5 for both. AI members on window edge: 133–143/300 |

| Date | Action Taken | One-Line Result |
|---|---|---|
| 10-15 | Round 2 Gate Segment: V1 ran through the gust pipeline on the 7 Step 1 events (~11.7 CU in this session, burn-rate × duration estimate) | Pipeline and runtime passed the gate: 173–312 s per group, median 22 s per member (vs. 55 s estimated), ~0.3–0.6 CU/group. Scientific result (seen sample, non-blind): wide-window AI/obs at 1–3 h: 0.51/0.59/0.62, HRRR GUST: 0.75/0.73/0.77; AI hits 0/3, 0/3, 0/2 (of HRRR GUST hits). Combined 10 events: median AI/HRRR GUST 1 h 0.84, 2 h 0.88, 3 h 1.01; ≥25.7 hits AI 3/2/2, HRRR GUST 6/6/4. Narrow window: 0/7 hits for both |
| 10-14 | Round 2 Segment ② V1/ERA5 (old cap ≤20 CU, counted toward this round; account balance drop of 18.03 CU as upper bound) | 153 checkpoint rows; 13 complete 10-member groups: 3 events (AMA, CDS, CID) × 1/2/3 h, 4 control groups (CDS 3 leads, CID L1); 9/10 controls cross-checked with neighboring METAR <40 kt, CNK has no neighboring station. Interim summary (not for verdict): median AI/obs 1 h 0.79, 2 h 0.83, 3 h 0.77, HRRR GUST 0.91/0.92/0.79; ≥25.7 hits AI 3/3, 2/3, 2/3, HRRR GUST 3/3, 3/3, 2/3; control false alarms 0 for both. Compute: ~7 new groups in this segment used ≤18 CU; at this pace the full ≥20+≥10 set would require >200 CU, so per-group runtime must be empirically verified in the gate segment |
| 10-14 | Round 2 Segment ① HRRRCast GPU compatibility (~1.1 CU) | Weights passed checksum again; isolated Python 3.10 + TensorFlow 2.15.1 recognized the A100, but the kernel repeatedly hung when loading `.keras`, so event inference did not run. Decision: defer HRRRCast (public version is 6 km, training period covers 2023, environment blocked); retry only if the V1 segment passes, using post-2024-04 events |

| Date | Action Taken | One-Line Result |
|---|---|---|
| 10-14 | Round 2 partial execution (pilot) | Two V1 cases (AMA 65 kt, CDS 62 kt) × 1/2/3 h × 10 members: gust proxy (AI 10 m wind × 1.6677, fitted on 2022) gave wide-window AI/obs 0.77–1.04, comparable to HRRR GUST 0.72–0.92, P(≥25.7) AMA 1.0, CDS 0.3–0.7; CONUS runs on A100 80GB (forward pass 97 s / 2 steps, 20.6 GB), one OKC 2026 case at L1 gave AI/obs 0.66, P=0; CONUS pressure fields only reach p20hl (~440 hPa); single-member smoke test with GFS conditioning passed; HRRRCast weights passed checksum but training period (2021-03 to 2024-03) covers 2023 events, preprocessing incomplete, no inference run. First session was reclaimed by the platform (pruned, reported to platform). None of the three criteria evaluated yet |
| 10-14 | HRRRCast usability verification | Public weights exist (~203 MB); however, the public version is 6 km, pressure levels only reach 200 hPa, GUST is a post-processed diagnostic (factor 1.4), weights have no separate license, and training period was unverified; not yet tested on Colab |
| 10-14 | Step 1 observation and baseline supplement: case-by-case peak check against IEM ASOS 1-min data; weight check on WoFSCast repository | In 6/7 cases, 1-min data matched the reported peak at the exact same minute and value; OFK had a missing observation at that exact minute (window max 53 kt, METAR PK WND recorded 56 kt), listed separately for review; same-source comparison cannot prove the absence of in-station QC peak removal. ≥25.7 m/s hits: AI 0/28, HRRR GUST 8/28, HRRR 10 m 1/28. WoFSCast repository has no weights or releases and its config points to internal paths → currently unavailable. Round 1 fail verdict unchanged |
| 10-14 | M: AWS AIWP (Pangu, GraphCast Operational) on 40 ASOS ≥50 kt thunderstorm high-wind days in 2023–24, 12–120 h lead times, verified against 13-level ERA5 with diurnally aligned persistence baseline and per-parameter criteria | DCAPE and 0–1 km wind shear passed across all five lead times; 0–6 km wind shear passed all five lead times for Pangu and up to 72 h for GraphCast; MUCAPE/MLCAPE failed across all five lead times for both models (median error often below persistence, but per-event win rate was only 14–25/40, short of 2/3). Verification time taken at nearest 00/12 UTC, up to 6 h off peak wind; verifies environment only, not gusts |

| Date | Action Taken | One-Line Result |
|---|---|---|
| 10-14 | Step 1: StormCast V1 (trained 2018-07 to 2021-12, Central US 512×640) on 7 ASOS thunderstorm high-wind events in 2023–24 × 1/2/3/6 h lead times | Failed preset thresholds: median AI/HRRR10 peak ratio 0.85 at 1 h, 0.61 at 2 h (1.11 at 3 h, 1.01 at 6 h); AI predicted ≥25.7 m/s in 0/28 cases; outperformed persistence in 6/7 events; ≤88 s per group, ~6 GPU hours for 100 events × 4 initializations. Framing issues: both AI and HRRR10 are 10 m mean winds, reaching only 0.3–0.5 against ASOS gusts whereas HRRR GUST reaches 0.75; single member; conditioning used retrospective ERA5. Soundings: only 14 model levels up to ~440 hPa, so CAPE cannot be used for L2. WoFSCast has no public weights/initial conditions, FuXi-Nowcast has no weights, CorrDiff weights are restricted |
| 10-14 | Deep Research verification (298d34) | Both reports agree: no convection-allowing AI model has undergone station-level verification for surface thunderstorm severe winds (literature absence); only StormCast is practically usable; DR misattributed the HRRR severe wind bias to Fovell 2022, whereas it is actually in Collins 2024 WAF PDF p. 10 (gust −11.32 m/s, cold-season downslope wind); ASOS QC erroneously removes ~10% of convective peaks (Zotero DZF9GFUU PDF p. 1); no precedent for "AI sounding → CM1." Sent for review |
| 10-14 | Grok rapid literature search + check of StormCast model card and FuXi-Nowcast abstract | Current StormCast weights trained 2018-07 to 2025-09, full CONUS 3 km, commercially usable under OpenMDW-1.1, has 10 m wind but no gust → verification events must be post-2025-10; FuXi-Nowcast has gusts but is limited to East China with no public weights; no precedent found for "AI sounding → CM1" |

# Parameter Source Table

| Parameter | Value | Source or "Assumption" | Places Checked |
|---|---|---|---|
| Verification neighborhood ±25 km / ±1 h | 25 km, 1 h | Assumption: ~8 grid points at 3 km covering convective position error; sensitivity: additionally report ±10 km / ±0.5 h | Pending rapid literature check |
| Severe wind threshold | 25.7 m/s (50 kt) | NWS severe wind standard | — |

# Open Items and Next Steps

- 10-16 01:52 Round 3 pre-registration frozen:
- Finalized, with SHA256 recorded.
- Based on the review verdict and decision record.
- Dispatched `wx` to execute S0+S1 (Colab, ≤12 CU).
- 10-16 02:10 `wx` returned pre-screening (~1.7 CU): tentative 295 V1 entries and 127 CONUS entries (grid boundary checks and control pairing not yet done, so not yet a frozen sample); S1 not started. At 02:20 dispatched follow-up: complete S0 wrap-up and proceed to S1, combined cap ≤12 CU across both orders.
- 10-16 02:28 returned: the two S0 orders were estimated at ~3.4 CU using the A100 rate, exceeding the 3 CU segment estimate, so `wx` stopped the instance; sample not frozen, S1 not started (grid check confirmed all tentative candidates lie inside the domain). Recorded as Deviation D1: segment CU figures are budget estimates only; S0 raised to ≤5 CU while keeping S0+S1 combined ≤12 CU. At 02:36 dispatched to complete both S0 freezing and S1 in one run.
- 10-16 03:26 returned, having stopped due to an accounting issue: the CPU session was billed at the A100 rate of 6.85 CU/h and stopped at an estimated ~9.8 CU. Qualified event pools stood at 301 for V1 and 132 for CONUS, with 120/80 provisionally sampled; control pairing was not yet finished. Recorded as Deviation D2: CPU sessions switched to billing at the empirically measured rate. At 03:36 dispatched to finish control pairing, sample freezing, and S1.
- 10-16 04:56 verification: pairing controls in the background on the CPU session, 22/200 completed with no missing controls so far; measured CPU burn rate is ~0.08 CU/h, confirming D2 accounting. `wx` scheduled a follow-up at 07:45 to freeze the sample and run S1. Left a note requiring S1(a) to use all native levels from HRRR `wrfnat`.
- 10-16 10:49 S0 frozen: `events_r3.csv` (…), `controls_r3.csv` (…), comprising 120 V1 events and 80 CONUS events with all 200 controls matched (qualified pool: 301 V1, 132 CONUS; switching to 35 dBZ added 0 events). 11:13 initial S1 results: (a) `1.113` / `r = 0.915`, (b) `1.00`, (c) failed (AI `Z1hl − terrain` spanned `−26` to `39 m`). Lead review uncovered a pressure unit bug in `θv` (Pa treated as hPa), voiding the initial run as D3; layer height was anchored to the lowest HRRR level as D4. Dispatched `wx` at 11:32 to rerun S1. Cumulative usage ~4.3 CU.

- 10-16 11:38 corrected S1 **passed all checks**: (a) formula vs operational `fg10m` median ratio `0.99999`, `r = 0.982` (10 events, 4,370 grid-point × time pairs); (b) using only StormCast's 14 levels yielded a peak median ratio of `1.000`; (c) after anchoring, 860 columns across two events had `NaN = 0` and 100% PBL top detection. Drift in AI `Z1hl` (`−26` to `39 m`) is anchored per D4; columns with non-monotonic layer heights are handled per D5. Total usage ~6.4 CU. Reported to parent Lead; **S2 (V1 120+120, ~144 CU) awaits parent project clearance**.
- (Original plan) Report to the parent Lead once S1 passes the gate. **Wait for parent project clearance before starting S2; do not launch directly.**
- 10-16 Parent node "AI Weather" approved the direction and cleared S0+S1 (~9 CU, conditional on finalizing after review). Dispatched `litwiki` to build literature cards (original text of Benjamin 2020 gust diagnostics, radar convective threshold), and sent out the final draft review (`--count 3`), specifically asking whether 80+80 in V1 provides sufficient statistical power and whether to raise it above 100 if more than 80 events are available. Once both return, issue a verdict, freeze the draft, and release S0+S1.
- `litwiki` returned (10-16), and I spot-checked the original text on PDF pp. 14 and 16 of Benjamin 2020, confirming accuracy. Two changes are required upon finalization: 1. Change the mixed-layer top criterion to θv(hl1, ~8 m) + 0.5 K (Benjamin 2020 PDF p. 14 / report p. 10), taking the maximum strictly within the PBL (PDF p. 16 / report p. 12). This eliminates the need for an assumption on `q2m`. 2. Keep the 40 dBZ radar threshold labeled as an "assumption": the literature supports >40 dBZ only for MRMS composite reflectivity (Liu 2021, Technical Note p. 2), and add >35 dBZ from Roberts & Rutledge 2003 (PDF p. 1 / journal p. 562) as a sensitivity analysis. Card identifiers: `H9ENWPG3_*`, `NR_liu_2021_*`. Report to the parent Lead once the S1 gate passes; **do not launch S2 directly**—it must be forwarded by the parent Lead to the parent project for clearance.
- 10-16 Round 3 pre-registration draft completed—**design only, zero compute spent**. L remains Shelved; whether to clear it is up to the parent node "AI Weather." Key points of the draft:
- Blind sample: V1 uses events from 2022, 2024, and 2025-01 to 09; CONUS uses nationwide post-2025-10 events; controls are "convective, non-severe-wind" cases paired 1:1 with events.
- Gusts switch to the HRRR mixed-layer diagnostic (Collins 2024 Eq. (1), journal p. 708), requiring reruns that save levels `hl1–hl20`.
- Primary lead time fixed at 2 h; three primary criteria: Brier score non-inferiority, hit/false-alarm non-inferiority (margin 0.15), and peak ratio ≥0.9.
- Sample size: V1 80 events + 80 controls, CONUS ≥50.
- Segments: S0 sampling freeze + S1 pilot + S2 V1 blind test (~105 CU); ~570 CU across all five segments; S2 and each subsequent segment require prior clearance from the parent Lead.
- Before finalization: assign Benjamin 2020 original text and radar threshold to `litwiki` for card creation; submit final draft for review under the frozen protocol.
- 10-15 12:50 Session B stopped by watchdog at 11:25 (46.19 CU, 230 rows, 20 complete groups); new session B2 exited immediately upon launch with `OptionalDependencyError`, producing no new results (reported to platform). At 12:58 dispatched `wx` to launch B3: install all optional dependencies matching original B and resume from row 230 to complete all 54 groups, with ~81 CU remaining in the round (cumulative ~134), stopping and reporting if the cap is reached.
- 10-15 Parent project approval: run CONUS Segment B to the full 54 groups with a whole-round cap of 215 CU, replacing "stop at 200"; resume within the same Segment B without recomputing or opening duplicate named sessions; stop and report to the parent Lead if exceeding 215 CU. Informed `ideas`.

Upon completion: if failed, send to 6 reviewers before shelving; if passed, report directly to the parent Lead. Scaling-up cost estimates (CONUS ~700 CU, V1 ~140 CU) are recorded for reference only and are not grounds for clearance.

- 10-15 10:42 Recalculation based on actual Segment B runtime (requested by parent Lead): Session B has run ~6.0 h, ~41 CU (estimated at 6.85 CU/h), completing 17/54 groups. Wall-clock time per group (1 event × 1 lead time × 10 members): ~14 min for 1 h, ~20.5 min for 2 h, ~27 min for 3 h (slower with more autoregressive steps), averaging ~20.5 min ≈ 2.3 CU/group—~1.6× slower than the 13 min/group used at 05:20 (those first 2 groups were 1 h leads). Round 2 cumulative usage is ~83 + 41 ≈ 124 CU, past the 120 CU reporting threshold (reported to parent Lead). The remaining 37 groups require ~87 CU, projecting ~210 CU for the full round, which will hit the 200 CU cap: stop at 200 CU and issue the evaluation on the ~49/54 completed groups under the frozen criteria while stating the exact sample size (events prioritized, leaving any gap in controls). The 05:20 statement "projected ~175, will not hit 200" is corrected per empirical measurements.
- Direct evidence for scaling up: CONUS takes ~20 min per group on a single A100, so 100 events × 3 lead times = 300 groups ≈ 100 GPU hours ≈ 700 CU; V1 takes 3–5 min per group, requiring ~20 GPU hours ≈ 140 CU at the same scale.
- 10-15 05:20 Round 2 compute accounting check (requested by parent Lead):
- By segment (in CU; "upper bound" = shared account balance drop including concurrent sessions; "est." = session duration × ~6.85 CU/h): pilot ≤26.8 (upper bound); Segment ① HRRRCast ~1.1 (est.); Segment ② V1 ≤18.0 (upper bound); Gate Segment ~11.7 (est.); Segment A ~25.7 (est.); Segment B through 05:10 ~3.3 (est.). Total ~86.6.
- Accounting basis: the older figure on the project page ("upper bound ~58") was the cumulative total prior to Segment A (≤46 + 11.7); the reported "~84/87" = 58 + Segment A 25.7 (+ Segment B 3.3). There was no double-counting or omission; the two figures correspond to different timestamps. The pilot and Segment ② figures are upper bounds whose true values are lower.
- Segment B projection: ~13 min/group, ~1.65 CU/group (the first 2 groups included package installation and weight loading, so actual per-group cost should be slightly lower), ~89 CU for 54 groups; 3.3 counted as of 05:10, leaving ~86 plus ~2–3 CU for installation/loading in one resumed session. Full round projected at ~175 CU, which will cross 120 (to be reported then) but will not hit 200. My 05:10 remark "will very likely hit the 200 cap" was a miscalculation and is retracted.
- Follow-up at 10:40 will recalculate based on actual per-group runtime in Segment B and record it here.
- 10-15 05:10 CONUS segment (Round 2 Segment B) running: 2/54 groups completed in ~27 minutes; CONUS is much slower per group than V1. Parent Lead instruction: upon reaching 50 CU in a single session, directly open a continuation session (resuming from checkpoints), keeping the full round ≤200 CU and reporting once past 120 CU; if unfinished at 200 CU, evaluate on the completed groups under the frozen criteria and state the sample count; report GPU minutes and CU per group, and estimate the cost for 100 events × 3 lead times.
- 10-15: Files >100 KB under `r2` (including `gate_7events` checkpoints) were moved to TACC `lab_archive` (manifest recorded) during Root Lead cleanup, to be verified and retrieved by the Root Lead once TACC recovers; continuation runs resume solely from the Colab local workspace, and if the evaluation lacks only those files, note "workstation checkpoints archived on TACC, not yet cross-checked." This explained the missing local files and was communicated to the platform.

- Round 2 cumulative Colab usage (upper bound, including concurrent sessions on the shared account): pilot ≤26.8 + Segment ① ~1.1 + Segment ② ≤18.0 ≈ ≤46 CU; report to parent Lead at 120 CU. Gate Segment passed (10-15 00:26); cumulative upper bound ~46 + 11.7 ≈ 58 CU. Cleared Segment A (V1 topped up to ≥20 events + ≥10 controls, ≤50 CU) and Segment B (CONUS 2026 events + ≥6 controls, ≤50 CU); hold the GFS segment until a model passes Criteria 1 and 2.
- `ideas` Round 2 (task brief, scheduled after M): gust conversion, 10-member ensemble, ≥20 events + ≥10 non-event controls, GFS real-forecast conditioning, and StormCast-CONUS trial run on post-2025-10 events with vertical level check; criteria fixed. 10-14 18:20 pre-check: estimated ~163 CU, halting at the 100 CU gate (666 candidate entries); Lead had previously cleared 200 CU, which the parent Lead amended on 10-14 to ≤20 CU per segment (200 CU pending parent project approval); dropped 6 h lead time, batched ensemble member inference, ordered as HRRRCast → V1 ERA5 → GFS conditioning → CONUS 2026 events.
- Parent Lead 10-14 added two diagnostic readouts (forwarded to `ideas`, criteria unchanged): report the fraction of maxima falling on the window edge (≥0.9× radius), and present narrow and wide windows side by side—in Round 1, ~20/28 AI wide-window maxima fell at 20–25 km, mostly reflecting background wind; the persistence baseline is too weak, so "6/7 outperforming persistence" does not count as evidence of skill, and conclusions rely on the two baselines HRRR GUST and "HRRR 10 m × same gust factor." Round 1 re-check: wide-window maxima on the window edge (≥22.5 km) occurred in 18/28 for AI, 13/28 for HRRR10, and 14/28 for HRRR GUST; in the narrow window (10 km / ±0.5 h), ≥25.7 m/s hits were 0/28 for all three.
- L2 soundings: StormCast V1 only reaches ~440 hPa and cannot feed CM1 directly; candidate alternatives are HRRR pressure-level products or the CONUS version (pending Round 2 vertical level check).
- 10-14 sounding supplement: CM1 column-extraction rules have been written as a testable script (2/2 synthetic tests passed); time offsets between the seven RAOBs and ASOS peaks range from 0.25 to 4.35 h (>3 h in 3 cases); true CM1 pre-convective inflow columns have not yet been extracted: they require case-by-case first ≥35 dBZ radar timestamps, HRRR reflectivity, and land masks, and raw profiles were not saved, so the existing `soundings.md` serves for reference only and cannot be used for L2 handoff. Decision: defer L2 column extraction and `input_sounding` preparation until Round 2 determines "worth continuing" (using native HRRR/HRRRCast levels + NEXRAD/MRMS); do not dispatch now.
- M completed (see table above): global AI models accurately predict wind shear and DCAPE 3–5 days ahead, whereas CAPE is unstable across 0.5–5 days. Implication for L2: use HRRR/ERA5 analyses to extract soundings for hindcasts; for forecasts, CAPE is the bottleneck, requiring reliance on short lead times from convection-allowing models (HRRRCast in Round 2 includes CAPE and 20 levels). No further additions to M.
- Grok rapid search verified (see table above); Deep Research verified and review verdict issued; HRRRCast added to Round 2; GitHub repository table provided by `ideas` in `models.md`.
- 10-14 response from the CM1 line: no duplication (they extract columns only from HRRR f00); column-extraction rules can be borrowed and have been forwarded to `ideas`; post-2022 cases with observed peaks are limited to NIP 2022-06-03 and ALTU 2023-07-03 (both within the current StormCast training period, so excluded as verification events); L2 small ensembles (10–20 members) will be run by the CM1 line once we deliver profiles in `input_sounding` format, scheduled after CRC group storage recovers and their sub-projects C/B launch.

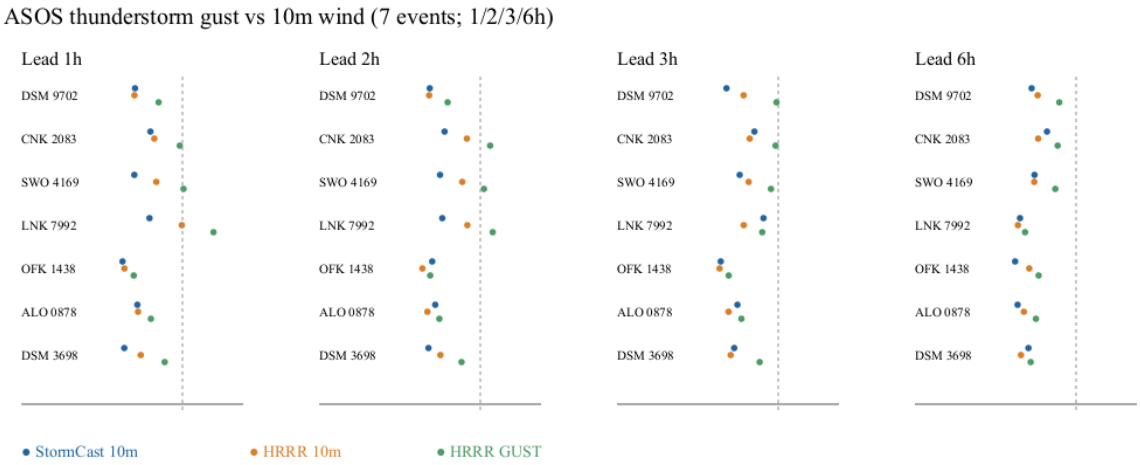
## Literature Basis

- PDF p. 3: 97 soundings driving CM1 downburst ensembles (L2 precedent)
- Near-domain precedents (cards not yet created): CGsNet Xiao et al. 2023 GMD 10.5194/gmd-16-3611-2023; Lagerquist et al. 2017 WAF 10.1175/WAF-D-17-0038.1; FuXi-Nowcast arXiv 2512.08974v2 (station gust verification, inferior to persistence at 13.9/17.2 m/s); HRRRCast arXiv 2507.05658; Cook 2023 Meteorology 10.3390/meteorology2020017; Collins et al. 2024 WAF (Zotero PU6IMCA9) PDF p. 10
- KYSQSJZT (Bolgiani 2020)
- Parent project AI Weather assessment table Row 1b: StormCast arXiv 2408.10958, WoFSCast 10.1029/2024GL112383, FuXi-Nowcast arXiv 2512.08974, Das 2024 10.1038/s41612-024-00834-8 (cards not yet created)

## Notes from Liang Hu

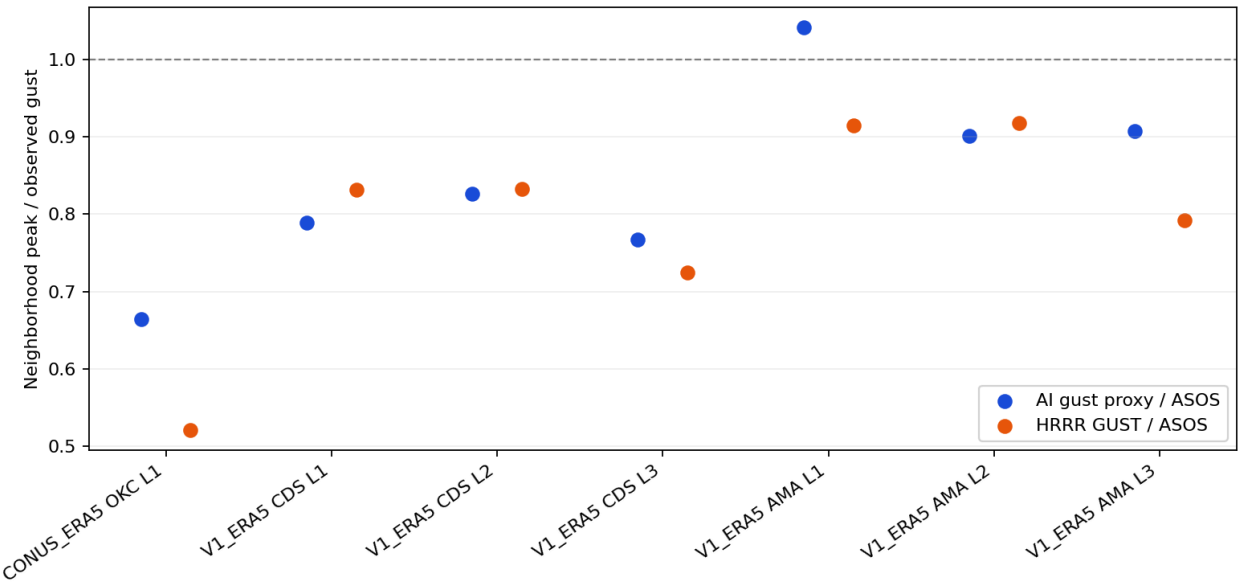
- 10-14: Once we reach the 3 km scale and capture favorable conditions, we can consider downscaling with CM1; precipitation is harder to forecast and is another hazard to consider later (P, not opening for now). The ultimate goal is large-scale forecasting and hindcasting; at this stage, make heavy use of Colab without CU limits.

# Key figures



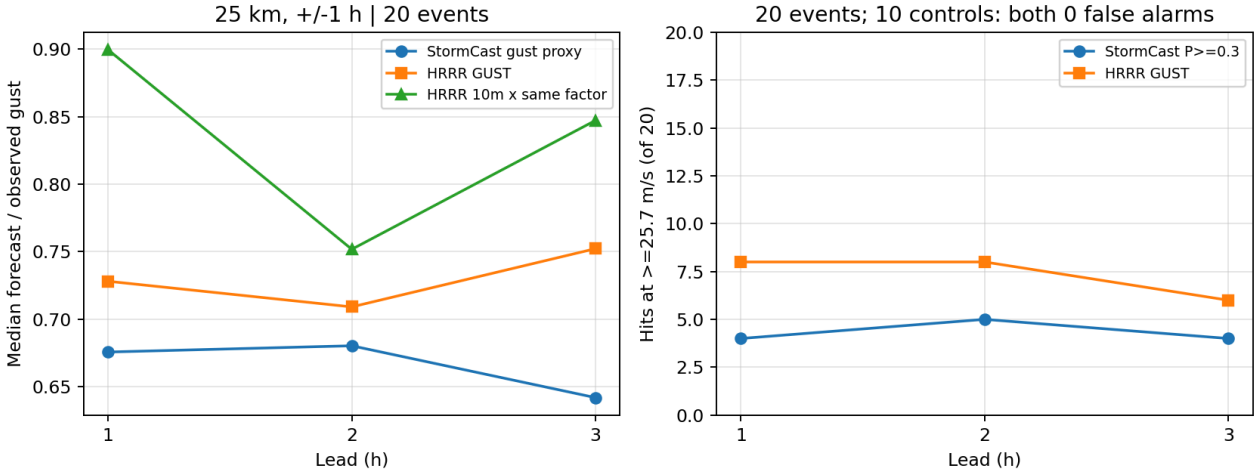

Sources: NOAA Storm Events; IEM ASOS; NOAA AWS HRRR; Copernicus ERA5/ARCO; NVIDIA StormCast V1. Dashed line = forecast/observed 1.0.

peak ratio




Pilot only (2 V1 events + 1 CONUS event); NOAA Storm Events/IEM ASOS, AWS HRRR, ARCO ERA5, NVIDIA StormCast. Not r2 gate.

pilot peak ratio




Sources: NOAA HRRR / Storm Events, IEM ASOS; NVIDIA StormCast V1; ARCO-ERA5.
Gust proxy = 10 m speed x 1.6677 (2022 fit); retrospective ERA5, not operational.

## v1 verdict